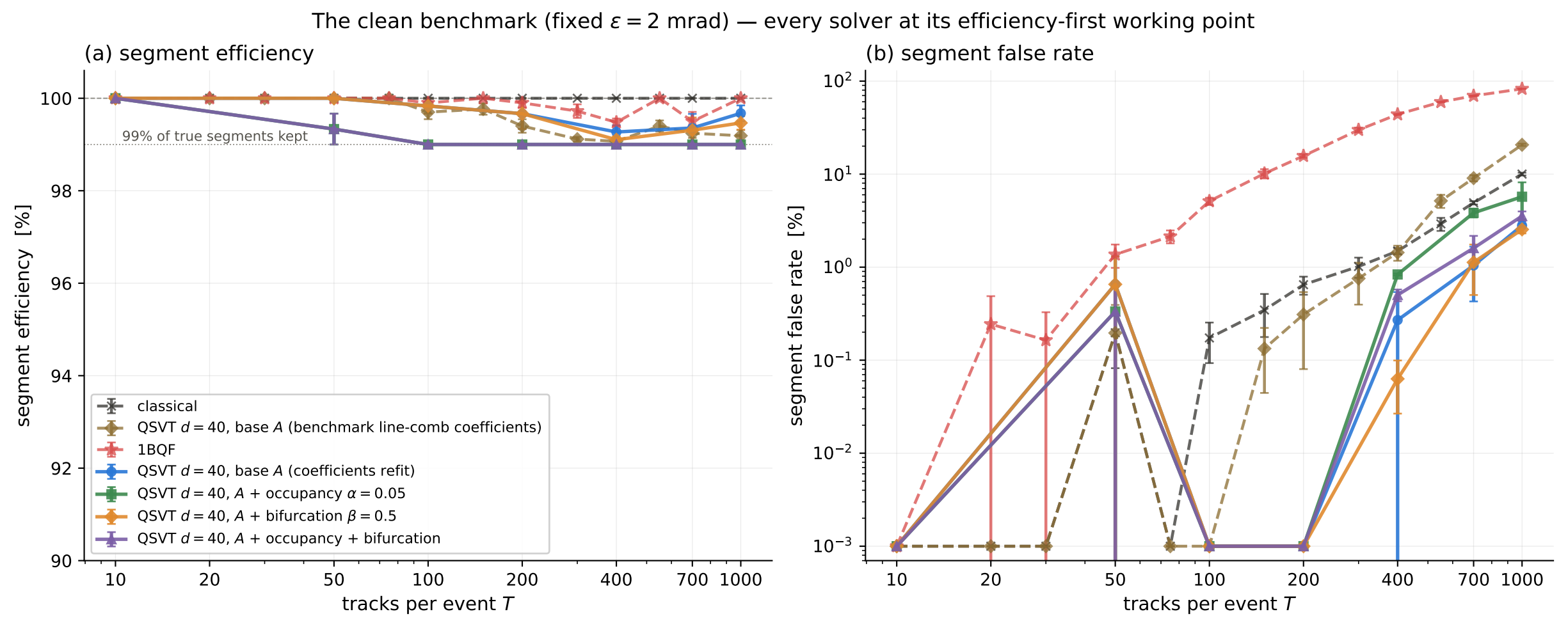


one configuration throughout: $\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 0$, no hit drop, $\varphi_{\max} = 0.2$, fixed $\varepsilon = 2$ mrad, $\gamma = 3$, $\delta = 1$, step kernel · benchmark campaign: mean±sem over event repetitions (classical 5–20, QSVT 5–10, 1BQF 3–10 per $T$) · refit series: the same campaign events, reps 0–2, mean±sem, QSVT coefficients least-squares fit per Hamiltonian (degree 40) · the threshold of every curve is set per event so that 99% of the true segments survive; panel (a) shows the convention holding, panel (b) carries the comparison · a measured 0% false rate is drawn at the $10^{-3}$ floor